A HANK Model for the Euro Area and Two Applications

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Introduction

- HANK models are now widely used at central banks
  - One of models at the ECB's Forecasting and Policy Modelling division: Ciccarelli et al, 2024
- Kase and Rigato (2025): first medium-scale HANK model used in FPM
  - Heterogeneous household block with information stickiness
  - Medium-scale DSGE: investment, price and wage stickiness etc.
  - Various aggregate shocks and policy instruments
  - Estimated on euro area data
- Ongoing work on extending our model along several dimensions
- Today:
  1. Current state of the model, focusing on the household block
  2. Two applications: household savings and fiscal scenarios following an energy shock

Model overview

- Heterogeneous household block with sticky information
- Nominal rigidities: price and wage stickiness
- Firm block with investment adjustment costs
- Financial frictions block with a financial accelerator mechanism and risk shocks
- Open economy with domestic goods, energy and non-energy imports
- Fiscal policy: government spending, tax rule, and multiple fiscal instruments
- Monetary policy: Taylor rule

Households

- Households choose consumption c_{it} and liquid asset holdings b_{it} to maximize:

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \exp\left(\sum_{k=0}^{s-1} \varepsilon_{t+k}^C\right) [u(c_{t+s}) - v(n_{t+s})]$$

- The budget constraint is

$$b_{it} + c_{it} = y_t e_{it} + (1 + r_t^b) b_{i,t-1} + T_{it}, \quad b_{it} \geq \underline{b}$$

- Disposable income (ex transfers) has a common component y_t and an idiosyncratic term e_{it}

$$y_{it} = (1 - \tau_t^\ell) w_t n_t + (1 - \tau_t^d) d_t$$

- Transfers T_{it} may depend on household characteristics depending on the application

Sticky information

- Households need to form expectations about y_t, r^b_t, ε^C_t, and T_{it}
- Assume sticky information as in Mankiw and Reis (2002) and Auclert, Rognlie and Straub (2020)
  - Each period, a fraction $1-\lambda$ of households update their information set
- Generates hump-shaped responses of consumption to shocks
- Important degree of freedom when estimating the model
  - Disciplines the time series behavior of C_t

Consumption bundles

- The final good is a CES bundle of energy C_{Et} (imported) and non-energy goods C_{Nt}

$$C_t = \left[\omega_E^{1/\eta_E} C_{Et}^{1-1/\eta_E} + (1-\omega_E)^{1/\eta_E} C_{Nt}^{1-1/\eta_E}\right]^{\eta_E/(\eta_E-1)}$$

- Non-energy goods are a CES bundle of domestic goods C_{Ht} and non-energy imports C_{Ft}

$$C_{Nt} = \left[\omega_F^{1/\eta} C_{Ft}^{1-1/\eta} + (1-\omega_F)^{1/\eta} C_{Ht}^{1-1/\eta}\right]^{\eta/(\eta-1)}$$

- Elasticity $\eta_E < \eta \rightarrow$ difficult to substitute away from energy
- Assume bundles are adjusted infrequently as in Auclert et al. (2024)
    - Each period, a fraction $1-\theta^C$ of households update their entire consumption bundle
- Analogous problem for investment and government consumption

Mutual fund

- Risk neutral, holds long-term government bonds, equity, and foreign bonds
- The total value of its assets is A_t evolves according to

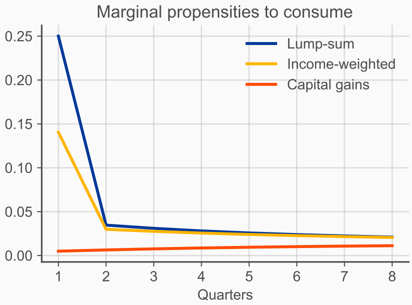
$$A_t = (1 + r_t^a)A_{t-1} - d_t$$

- r_t^a is the total return on its portfolio and d_t is the dividend paid to households
- Dividends are determined by the following rule:

$$d_t = d + \chi\left[(1 + r_t^a)A_{t-1} - (1 + r^a)A\right]$$

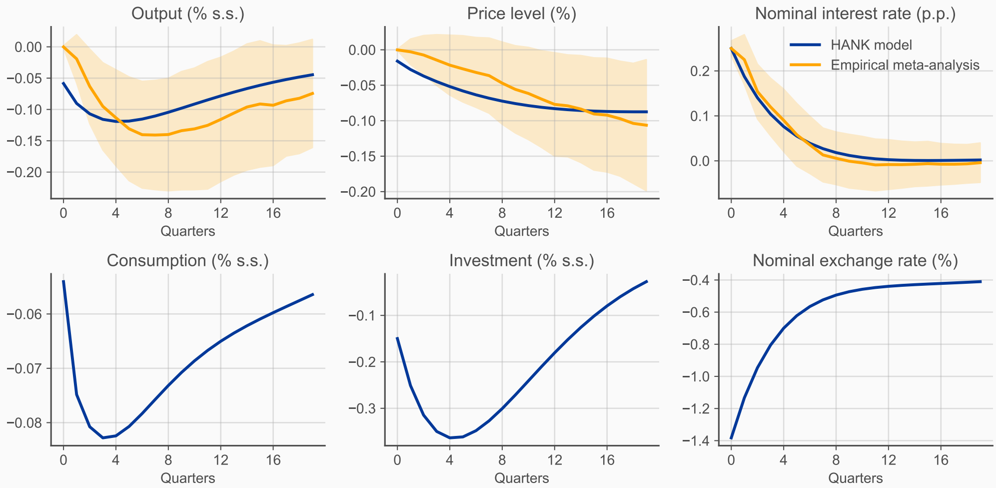
- The parameter χ governs how fast consumption responds to changes in financial wealth

Marginal propensities to consume



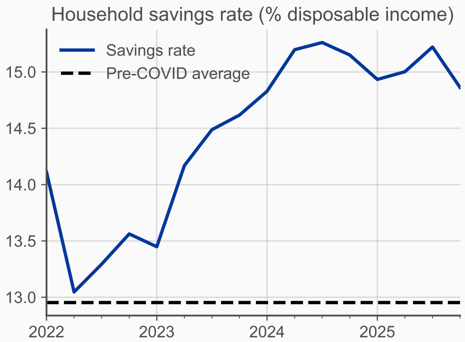
- Target an average MPC of 0.25 and $r = 1\%$ p.a.
  - Requires $B = 70\%$ of annual GDP, $\beta = 0.992$
  - Results in an income-weighted MPC of 0.14
- Calibrate χ to target annual MPCs out of capital gains
  - Chodorow-Reich, Nenov and Simsek (2021) find 0.032
  - Our model gives 0.028 and 0.042 over the first two years
  - This requires $\chi = 0.035$ (for $\lambda = 0.92$)

Monetary shock IRFs



Applications

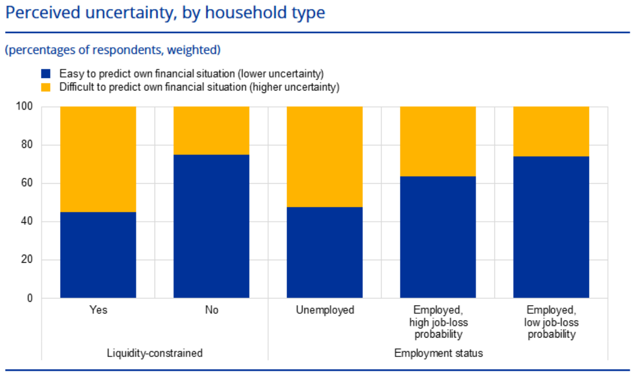
Application 1: household savings



- Household savings substantially higher than pre-pandemic levels
- Potential explanations (Bobasu, Gareis and Stoevsky, 2024):
  - Increased uncertainty
  - Tight labor markets and higher wages
  - Need to rebuild real wealth after the inflation spike
  - Higher interest rates

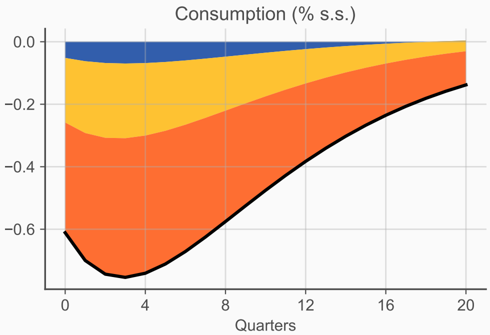
Substantial heterogeneity in perceived uncertainty

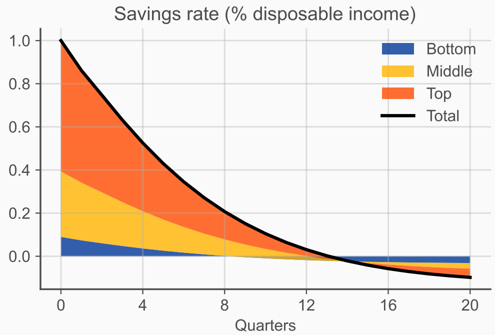
- ECB Consumer Expectations Survey results from Dimou, Dossche, Hütten and Kocharkoc (2026)



- Can uncertainty at the bottom of the income distribution be a risk to GDP growth?
  - More generally, can microdata be informative about macroeconomic risks?

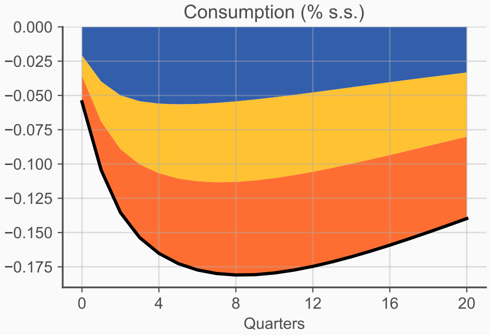
Can stronger saving at the bottom affect aggregate consumption?

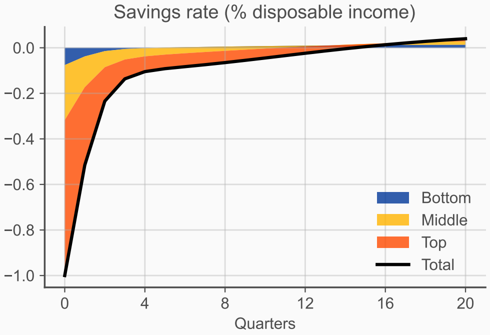




- Response of consumption to a discount factor shock decomposed into income terciles
  - Not exactly the same as an uncertainty shock, but works as a proxy
- Stronger saving incentives at the bottom have limited effects on aggregate consumption
- The bottom of the income distribution accounts for a small share of consumption

But some shocks propagate strongly through the bottom of the distribution: energy price shock



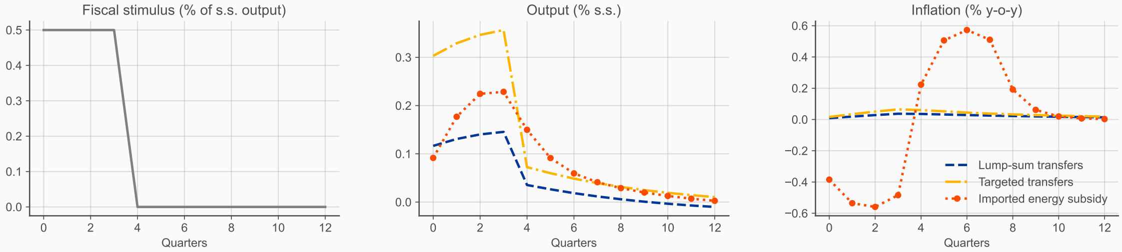


- Response of consumption to an increase in imported energy prices
  - Higher energy prices $\longrightarrow$ lower disposable income $\longrightarrow$ operates via MPCs
- Strong contribution coming from the bottom tercile: low consumption share, but high MPCs
- Savings are still driven by the top
- More details in Battistini, Bobasu, Kase and Rigato (2026)

Application 2: fiscal policy scenarios following an energy shock

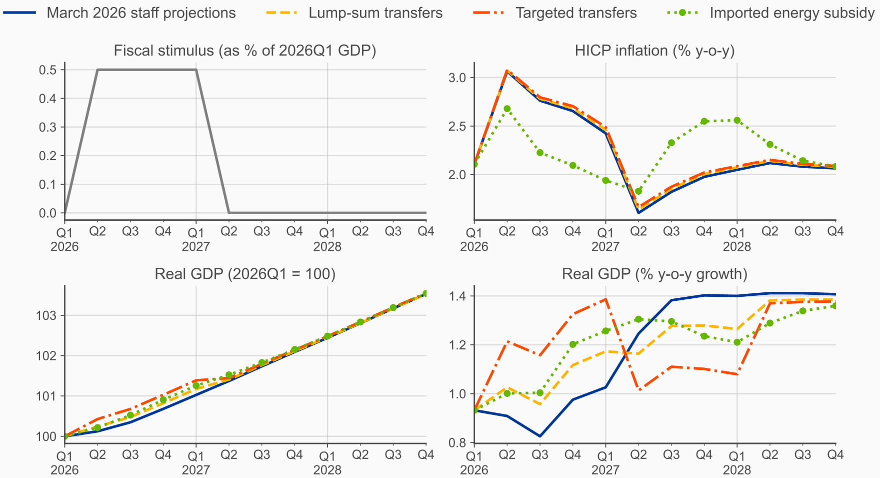
- Large increase in oil and energy prices in the past few months
- In the last energy crisis, governments responded with a mix of fiscal instruments
- How effective are these instruments and how do they affect real GDP and inflation projections?
- Compare illustrative fiscal policy scenarios based on different instruments
  - Targeted transfers to households (transfer to the lower tercile)
  - Untargeted (lump-sum) transfers
  - Subsidies on imported energy

Effects of fiscal policy on GDP and inflation



- Fiscal multipliers: targeted transfers > energy subsidies > untargeted transfers
  - Energy subsidies also stimulate investment
- Muted effects on inflation for transfers: flat Phillips curve
- Mechanical effect on energy subsidies, but also second-round effects

Different fiscal policy scenarios



Takeaways

- Substantial progress in HANK modelling at FPM over the past few years
- What this framework lets us do:
  - Identify macroeconomic risks using microdata
  - Assess the effectiveness of different policy instruments
  - Different transmission channels relative to other models
  - Inequality matters in its own right
- We welcome your comments and suggestions!
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